

$$A = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

For Cholesky Decompositⁿ

$$A = LL^T$$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

$$l_{11} = \sqrt{a_{11}} = \sqrt{9} = 3$$

$$l_{21} = \frac{15}{3} = 5$$

$$\Rightarrow l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{50 - 25} = 5$$

$$\Rightarrow |l_{22}| = 5$$